AP PHYSICS 1

Unit 1 • Worksheet 1.2A • Answer key

Worksheet 1.2A • Displacement, Velocity, and Acceleration

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A runner covers 100 m in a straight line in 12.5 s. What is her average speed?

(A) $12.5\,\text{m/s}$
(B) $80\,\text{m/s}$
(C) $0.125\,\text{m/s}$
(D) $8\,\text{m/s}$

Solution

Answer: (D). $100/12.5 = 8\,\text{m/s}$.

Question 2

A car speeds up from 10 m/s to 25 m/s in 5 s. What is its average acceleration?

(A) $2\,\text{m/s}^2$
(B) $5\,\text{m/s}^2$
(C) $3\,\text{m/s}^2$
(D) $7\,\text{m/s}^2$

Solution

Answer: (C). $a = \Delta v/\Delta t = (25-10)/5 = 3\,\text{m/s}^2$.

Question 3

An object has velocity $v = -6\,\text{m/s}$ and acceleration $a = +2\,\text{m/s}^2$. What is happening?

(A) It is speeding up in the negative direction.
(B) It is slowing down.
(C) It is at rest.
(D) It is speeding up in the positive direction.

Solution

Answer: (B). v and a have opposite signs, so the object slows down.

Question 4

An object starts from rest with a constant acceleration of $3\,\text{m/s}^2$. What is its speed after $4\,\text{s}$?

(A) $7\,\text{m/s}$
(B) $12\,\text{m/s}$
(C) $24\,\text{m/s}$
(D) $0.75\,\text{m/s}$

Solution
Answer: (B). $v = v_0 + at = 0 + 3(4) = 12\,\text{m/s}$.

Question 5

For the object in the previous question, how far does it travel in those $4\,\text{s}$?

(A) $12\,\text{m}$
(B) $48\,\text{m}$
(C) $24\,\text{m}$
(D) $6\,\text{m}$

Solution
Answer: (C). $\Delta x = \frac{1}{2}at^2 = \frac{1}{2}(3)(16) = 24\,\text{m}$.

Question 6

A car moving at $30\,\text{m/s}$ brakes with an acceleration of magnitude $6\,\text{m/s}^2$. How far does it travel before stopping?

(A) $75\,\text{m}$
(B) $150\,\text{m}$
(C) $5\,\text{m}$
(D) $90\,\text{m}$

Solution
Answer: (A). $0 = 30^2 - 2(6)\Delta x$, so $\Delta x = 900/12 = 75\,\text{m}$.

Question 7

A ball is thrown straight up at $15\,\text{m/s}$. How long does it take to reach its highest point?

(A) $3\,\text{s}$
(B) $0.67\,\text{s}$
(C) $1.5\,\text{s}$
(D) $15\,\text{s}$

Solution
Answer: (C). At the top $v = 0$: $0 = 15 - 10t$, so $t = 1.5\,\text{s}$.

Question 8

At the highest point of its path, a ball thrown straight up has

(A) $v = 0$ and $a = 0$
(B) $v = 0$ and $a = 10\,\text{m/s}^2$ downward
(C) $v = 10\,\text{m/s}$ upward and $a = 0$
(D) $v = 0$ and $a = 10\,\text{m/s}^2$ upward

Solution
Answer: (B). The velocity is zero for an instant, but gravity still acts: $a = g$ downward at every point.

Question 9 (Free response)

A train starts from rest and accelerates at $0.50\,\text{m/s}^2$ for $40\,\text{s}$. It then brakes with an acceleration of magnitude $1.0\,\text{m/s}^2$ until it stops.

(a) Find its speed at $t = 40\,\text{s}$.
(b) Find the distance traveled during the first $40\,\text{s}$.
(c) Find the braking distance and the braking time.

Solution
(a) $v = 0.5(40) = 20\,\text{m/s}$. (b) $\Delta x = \frac{1}{2}(0.5)(40)^2 = 400\,\text{m}$. (c) $0 = 20^2 - 2(1.0)\Delta x \Rightarrow \Delta x = 200\,\text{m}$; $t = 20/1.0 = 20\,\text{s}$.